# Salient Characteristics / Item Specifications
## Steam Manhole Cooling Ventilation Trailer

### 1.0. Background

The National Institute of Standard and Technology (NIST) campus located in Gaithersburg, MD experienced a total loss of site steam delivery in 2023. Following the total loss of site steam delivery, NIST hired a contractor to conduct the necessary steam manhole (SMH) repairs, which included the utilization of a specialized steam manhole ventilation trailer unit.

The steam supply maintains a temperature of **372°F**. When fully insulated, the Steam Manhole (SMH) piping remains at a constant temperature of approximately **140°F**. However, SMHs experiencing steam or condensate leakage can exceed temperatures of **180°F to 190°F** in the confined space.

These elevated temperatures are difficult to effectively mitigate using traditional confined space blower units, regardless of the number of units utilized. Because ambient air temperature is the primary factor in cooling effectiveness, a specialized trailer ventilation unit was utilized and found to successfully reduce the internal SMH temperatures to near-ambient levels during summer repair efforts.

### 2.0. Purpose

The primary objective of this requirement is for a Contractor to provide a mobile cooling solution that reduces and maintains internal SMH temperatures at near-ambient levels. This cooling is critical for the health and safety of NIST work crews during prolonged maintenance and repair services, which often span multiple consecutive workdays.

The core function of this unit is to house a 5-ton HVAC unit that supplies cold air into the SMH space.

The purchase of this cooling unit will allow NIST Pipe and Electric Shop personnel to work in both steam and electrical manholes safely without risk of the high temperatures.

### 3.0. General Requirements / Specification

The Contractor shall provide a self-contained, trailer mounted, cooling unit that meets the requirements and specifications detailed herein.

All items shall be new. Prototypes, demonstration models, used or refurbished equipment will not be accepted.

### 3.1. Trailer

**Trailer basis of Design**: Carry-On Trailer Model 7X12GWHS16 **or similar**

The trailer consists of the following components:

- 7’ x 12’ Bumper Pull Utility Trailer
- DOT Over-the-road certified with VIN and transferable title
- 2” ball coupler hitch
- GVWR; 2,990 Pounds
- 15” Pneumatic Tires
- Mesh metal “Ramp” Gate

- Wood Deck

**3.2. Fuel Transfer Tank**

**Basis of Design:** Better Built Model 720467115849 **or similar**

- Black: powder Coated
- 36-gallon, steel transfer tank
- Internal Baffles
- 2” reinforced bungs
- Constructed of 14 ga. steel

**3.3. HVAC Unit**

**Basis of Design:** Carrier 5-ton, 50NT-B Comfort 13.4 SEER2 Single-Packaged Heat Pump System **or similar**

5 Ton Packaged Heat Pump

**3.4. Diesel Generator**

**Basis of Design:** Kubota GL14000 **or similar**

- Diesel engine
- Generator and engine shall be directly coupled together
- Rated Voltage 240 / 120
- Prime Output 12 kVA
- Standby Output 14 kVA
- Frequency: 60 HZ
- Phase: Single
- Receptacles 120V-20A GFCI x 2 (4 receptacles) 120/240V-50A Twist type x 1
- One-side maintenance access
- Automatic engine shutdown at water temperature rise or oil pressure drop
- Double element air cleaners

**3.5. Ventilation Duct (1 each)**

**Basis of Design: CoolFlex Ventilation Duct or similar**

- Laminated PVC ventilation duct
- Yellow, 14 Ounce material
- Support wire – steel
- 25 feet long
- Carry case

**3.6. Stabilizing Jack**

**Basis of Design:** Pro Series Round, Pipe-Mount Swivel Jack w/ Footplate - Sidewind - 10" Lift -5,000 lbs., Part Number: PS1401420303 **or similar**

- Round
- Pipe-Mount Swivel Jack w/ Footplate - Sidewind - 10" Lift

### 3.7. Utility Storage Box

- 6' – 8' Long
- Steel
- Black Powder Coated
- Lockable lid

### 3.8. Rubber Wheel Chock

- One Set (two wheel chocks) required
- Medium duty ratings (for up to 20-ton vehicles)
- Constructed from industrial strength thermoplastic rubber

### 3.9. Fuel

Fuel tank will be full at delivery

### 4.0. Design & Build Requirements

The trailer design and build shall incorporate the following specifications:

- Trailer shall be 7' wide x 12' long.
- HVAC unit and generator shall be raised off the trailer deck by 1 – 1-1/2" to allow full air circulation under the equipment.
- Trailer rails shall be modified to allow for full access to HVAC unit and generator maintenance points.
  - Equipment box placement shall not impede this access.
- HVAC unit air discharge and filter access shall be fully accessible.
- Install ball valve fuel shut off at tank discharge.
- Fuel lines and fuel filter(s) from fuel tank to generator shall be fully protected and supported across the entire run.
- Install welded brackets for wheel chock storage off trailer frame (one each side).
- Install sidewind stabilization jack on each rear corner of trailer.

### 5.0. Tasks

Contractor shall:

1. **Verification:** It is the responsibility for the Contractor to independently verify all conditions, including but not limited to specified cooling capacity, and trailer towing/footprint constraints to ensure the proposed unit meets all performance specifications.
2. **Design & Layout Approval:** Provide the COR with the final layout of all major components for approval prior to permanently attaching equipment to trailer.
3. **Procurement & Integrated Assembly:** Supply all labor, materials, and equipment identified in the approved design to produce the complete trailer unit. This includes all mounting hardware, electrical wiring, and weatherproofing.
4. **Pre-Delivery Inspection:** Notify COR upon completion of final assembly to arrange a functional test and pre-delivery inspection.
5. **Final Delivery:** Deliver the fully commissioned and operational trailer to NIST Gaithersburg.

6. **Technical Documentation:** Provide all equipment information, manuals, cut sheets, and titles / warranties with trailer delivery.
7. **Maintenance Support:** Supply a 1-year inventory of all OEM-recommended preventive maintenance consumable parts for both the HVAC unit and the generator.
8. **Onsite Training:** Provide up to 4 hours of onsite training with NIST personnel.

**6.0. Delivery**

Delivery of the completed trailer shall occur no later than 90-days after award of the contract.

- The Contractor shall deliver the completed trailer to NIST, 100 Bureau Dr., Building 301, Gaithersburg, MD 20899-1640
- Contractor will be required to enter site through Gate “F” and then go through Security inspection.

**7.0. Inspection and Acceptance Criteria**

Final acceptance of the equipment will be conducted once received and inspected for damage and operation in accordance with FAR Clause 52.212-4(b). Any damage, repairs, or re-work shall be done at no cost to the Government.

**8.0. Warranty**

The Contractor shall provide a one-year warranty will be provided on workmanship.

The Contractor shall provide a one-year warranty of all equipment from the date of delivery should the original manufacturers’ warranty be voided in any way by installation, alterations, or alternative use provisions.